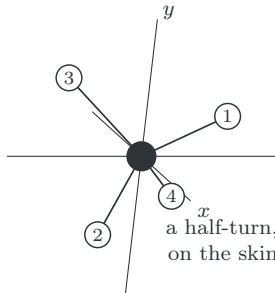
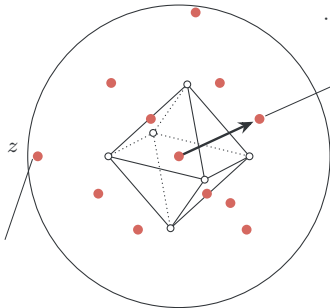


$SO(3)$  as a ball: direction the axis, distance the angle



$\mathbf{X}_0$  in its body frame



the twelve rotations of  $T$ , one cell  $\mathcal{U}$ ,  
and the lift  $r \rightarrow r\mathbf{R}_g$  of the loop

... twelve sheets, one per rotation of  $T$

the third-turn  
about bond 1

